Autoregressive Conditional Heteroskedasticity and Changes in Regime

I. Introduction and Motivation

A. Importance of Volatility Modeling

Modeling how financial market volatility changes over time is crucial for two primary reasons:

1. **Asset Pricing:** The riskiness of an asset is an important determinant of its price. Indeed, empirical estimation of the conditional variance of asset returns forms the core of several hundred studies of financial markets.

2. **Econometric Efficiency:** Efficient econometric inference about the conditional mean of a variable requires correct specification of its conditional variance.

B. Limitations of Traditional ARCH/GARCH Models

The popular traditional approach to volatility modeling is the ARCH specification introduced by Engle (1982), or the Generalized ARCH (GARCH) specification introduced by Bollerslev (1986).

1. **Persistence vs. Forecasting:** ARCH models often impute a lot of persistence to stock volatility and yet give relatively poor forecasts. The Student t GARCH-L(1, 1) model actually yielded poorer forecasts (MSE = 815.6) than just using a constant variance (MSE = 757.9), resulting in an R-squared of -0.08.

2. **Structural Change Hypothesis:** The high estimated value for the persistence parameter λ (estimated to be 0.96 for GARCH-L(1, 1)) may reflect structural changes that occurred during the sample in the variance process. This high persistence implies that any change in the stock market volatility this week continues to have nonnegligible consequences a full year later: $(0.96)^{52} = 0.12$. The model's forecasts continue to worsen relative to a constant-variance specification as the forecast horizon increases.

3. **Nature of Shocks:** Extremely large shocks, such as the October 1987 crash, might arise from quite different causes and have different consequences for subsequent volatility than do small shocks.

C. Data Used

The stock price series used is the value-weighted portfolio of stocks traded on the New York Stock Exchange contained in the CRISP data tapes. The raw data are returns from Wednesday of one week to Tuesday of the following week. The sample starts with the week ended Tuesday, July 3, 1962, and ends with the week ended Tuesday, December 29, 1987. Estimation began with the week ended Tuesday, July 31, 1962. Thus $\mathbf{T = 1327}$ observations were used to estimate each model. Let y_t denote the weekly stock return measured in percent. In all investigated models, the process u_t described by the ARCH process is the residual from a first-order autoregression for stock returns: $y_t = \lambda + \phi y_{t-1} + u_t$.

II. Traditional Model Refinements and Persistence

A. Best Traditional Specification (Student t GARCH-L(1, 1))

The GARCH(1, 1) specification is the most popular ARCH model used to describe financial market volatility. Modifications made to improve the model's ability to describe the data included:

1. **Student t Distribution:** The residual u_t is treated as drawn from a Student t distribution with ν degrees of freedom (normalized to have unit variance), following a suggestion by Bollerslev (1987) and Baillie and DeGennaro (1990).

2. **Leverage Effect:** A leverage effect, predicting $\zeta > 0$, is included, where d_{t-1} is a dummy variable equal to unity if $u_{t-1} \leq 0$ and zero if $u_{t-1} > 0$. The variance equation takes the form:

$$\sigma_t^2 = \alpha_0 + \alpha_1 u_{t-1}^2 + \zeta d_{t-1} u_{t-1}^2 + h_1 \sigma_{t-1}^2$$

The maximum likelihood estimates for this specification were:

$$y_t = 0.31 + 0.27y_{t-1} + u_t$$

$$\sigma_t^2 = 0.28 + 0.07u_{t-1}^2 + 0.23d_{t-1}u_{t-1}^2 + 0.78\sigma_{t-1}^2$$

The estimated degrees of freedom were $\hat{\nu} = 5.6$. All of the parameters were highly statistically significant.

B. High Persistence Calculation

The persistence is measured by the decay parameter λ, derived from the difference equation governing the m-period-ahead forecast of the variance $E(u_{t+m}^2|u_t, u_{t-1}, \ldots)$:

$$\lambda = (\alpha_1 + h_1 + \zeta/2)$$

Based on the estimated parameters, $\hat{\lambda}$ is estimated to be **0.96**. This implies **extremely persistent movements** in stock price volatility. This consequence is seen dramatically in the effects of the October 1987 crash, where the model anticipates possible weekly stock price movements in excess of 10% through the end of 1987. The GARCH confidence intervals shrink very gradually in response to these large outliers.

III. Markov-Switching ARCH (SWARCH) Methodology

A. Regime Structure

The SWARCH model explores a specification where the parameters of an ARCH process can occasionally change.

- **Unobserved State:** The unobserved random variable s_t can take on values $1, 2, \ldots, K$.
- **Markov Chain:** s_t is described by a Markov chain, where the transition probabilities are $P_{ij} = \text{Prob}(s_t = j|s_{t-1} = i)$. The transition probabilities are collected in a $(K \times K)$ matrix P. Each column of P sums to unity.
- **Parsimonious Parameterization:** The idea is to model changes in regime as changes in the scale of the process.
- **Residual Definition:** The residual u_t is modeled as a scaled version of an underlying ARCH-L(q) process $\tilde{u}_t$:
$$u_t = \gamma_{s_t}\tilde{u}_t$$
- **Scaling Factors:** γ_{s_t} is the constant factor dependent on the state s_t. The factor for the first state, γ_1, is normalized at unity, with $\gamma_j > 1$ for $j = 2, 3, \ldots, K$.
- **Underlying ARCH Process:** $\tilde{u}_t = \tilde{h}_t\tilde{\epsilon}_t$, where $\tilde{\epsilon}_t$ is i.i.d. zero mean, unit variance sequence. $\tilde{h}_t^2$ obeys:
$$\tilde{h}_t^2 = \alpha_0 + \alpha_1\tilde{u}_{t-1}^2 + \alpha_2\tilde{u}_{t-2}^2 + \cdots + \alpha_q\tilde{u}_{t-q}^2 + \zeta d_{t-1}\tilde{u}_{t-1}^2$$
- **Conditional Variance:** Conditional on the regime history, the variance implied for u_t is $E(u_t^2|s_t, s_{t-1}, \ldots, s_{t-q}, \ldots)$ $\gamma_{s_t}^2\tilde{h}_t^2(s_t, s_{t-1}, \ldots, s_{t-q})$.

B. Inference and Estimation

The methods developed in Hamilton (1989) can be used to evaluate the sample log-likelihood function: $\mathcal{L} = \sum_{t=1}^{T} \ln f(y_t|y_{t-1}, y_{t-2}, \ldots, y_{-3})$.

1. **Filter Probability:** This inference about the state based on information observed through date t is called the filter probability: $P(s_t, s_{t-1}, \ldots, s_{t-q}|y_t, y_{t-1}, \ldots, y_{-3})$. There are K^{q+1} possible configurations for $(s_t, s_{t-1}, \ldots, s_{t-q})$.
2. **Smoothed Probability:** This uses the full sample of observations (up to date T) to construct the smoothed probability $P(s_t|y_T, y_{T-1}, \ldots, y_{-3})$. There are K separate numbers for each date t in the sample.

C. Forecasting and Persistence in SWARCH

The model provides a framework that could generate nonlinearity in stock return volatility documented by Friedman and Laibson (1989) and Friedman (1992).

1. m**-Period-Ahead Forecast** $\sigma^2_{t+m|t}$**:** This is the conditional expectation of u^2_{t+m}:

$$\sigma^2_{t+m|t} = E(u^2_{t+m}|y_t, y_{t-1}, \ldots, y_{-3})$$

   It is calculated by weighting each conditional forecast by the filter probability of that configuration:

$$\sigma^2_{t+m|t} = \sum_{s_t=1}^{K} \cdots \sum_{s_{t-q+1}=1}^{K} h(\cdot) \times P(s_t, s_{t-1}, \ldots, s_{t-q+1}|y_t, y_{t-1}, \ldots, y_{-3})$$

   The forecast conditional on a known state path s is $h(\cdot)$, which equals $\gamma^2_{s_{t+m}} \cdot \tilde{h}^2_{t+m|t}(\ldots)$.

2. **ARCH Component Persistence (**λ**):** The underlying ARCH component forecast $\tilde{h}^2_{t+m|t}$ obeys a q-th order difference equation for $m \geq 2$:

$$\tilde{h}^2_{t+m|t} = \alpha_0 + (\alpha_1 + \zeta/2)\tilde{h}^2_{t+m-1|t} + \alpha_2\tilde{h}^2_{t+m-2|t} + \cdots + \alpha_q\tilde{h}^2_{t+m-q|t}$$

   The persistence of the ARCH component is measured by the **largest eigenvalue** (λ) of the companion matrix associated with this difference equation.

IV. Empirical Results and Model Comparison

A. Model Selection and General Findings

- **Innovations Distribution:** Wald tests and likelihood ratio tests overwhelmingly rejected the Normal in favor of the **Student** t **formulation**. The third ARCH parameter (α_3) was not statistically significant.
- **Focus:** Tables 1 through 3 primarily report the results from **SWARCH-L(**K**, 2) specifications driven by Student** t **innovations**.
- **AIC/Schwarz:** Based on Akaike's criterion (AIC), the **SWARCH-L(4, 2)** is the best model. Based on Schwarz's criterion, the GARCH-L(1, 1) is best, followed by the SWARCH-L(3, 2).

B. Results of Student t SWARCH-L(3, 2)

- **Mean Equation:** $y_t = 0.35 + 0.25y_{t-1} + u_t$. The autoregressive coefficient $\phi = 0.25$ is clearly nonzero, confirming positive serial correlation.
- **Degrees of Freedom (**ν**):** Estimated at 7.2.
- **ARCH Component** $\tilde{h}^2_t$**:** $\tilde{h}^2_t = 0.57 + 0.03\tilde{u}^2_{t-1} + 0.12\tilde{u}^2_{t-2} + 0.42d_{t-1}\tilde{u}^2_{t-1}$.
- **Volatility Scale:** $\gamma_1 = 1$, $\gamma_2 = 4.4$, $\gamma_3 = 13.1$. State 2 is four times as great as State 1, while State 3 is thirteen times as large.
- **Regime Persistence:** The states are highly persistent. State 1 would be expected to last 132 weeks; State 2, 116 weeks; and State 3, 59 weeks.
- **ARCH Persistence:** The underlying ARCH-L(2) process persistence is much less so, with $\hat{\lambda}$ estimated to be **0.48**. This means volatility effects captured by $\tilde{u}_t$ die out almost completely after about a month ($\lambda^4 = 0.05$).
- **Recession Link:** High-volatility episodes (State 3) are related to general business downturns, such as the 1969–70 and 1973–75 recessions.
- **Outlier Response:** The SWARCH confidence intervals shrink **quickly** in response to large outliers, returning to the trend level associated with a particular regime, which contrasts sharply with the slow decay implied by GARCH. This ability to track year-long shifts without imputing high persistence to individual outliers explains the improvement in forecasting.

C. Results of Student t SWARCH-L(4, 2) (Extreme State)

- **Mean and ARCH parameters:** The mean and ARCH parameters are similar to the 3-state model. $\hat{\nu} = 8.7$.
- **Volatility Scale:** $\gamma_1 = 1$, $\gamma_2 = 4.5$, $\gamma_3 = 13.8$, $\gamma_4 = 169$. The fourth state corresponds to an order of magnitude increase in the variance even beyond State 3.
- **Observations in State 4:** Only **two observations** are clearly generated by state 4: the **October 1987 crash** and a 5.8% surge in stock prices in the first week of January 1963.
- **Transition Insight:** p_{34} is estimated to be zero. The extreme State 4 appears never to have developed from a high-volatility State 3 but rather **always follows the moderate-volatility State 2**.
- **Numerical Caveats:** The SWARCH-L(4, 2) specification **proved extremely difficult to maximize**, owing in part to a **nearly singular Hessian**. As a result, "asymptotic" standard errors for some parameters **are not meaningful**.

V. Model Comparison and Conclusion

A. Loss Functions and Forecasting

Table 1: Model Comparison based on Loss Functions

Model	MSE	MAE	Selection Criterion
Constant Variance	757.9	7.30	N/A
Student t GARCH-L(1, 1)	815.6 (Improvement -0.08)	7.05	3rd (by Schwarz)
Student t SWARCH-L(3, 2)	814.3 (Improvement -0.07)	6.53	2nd (by Schwarz)
Student t SWARCH-L(4, 2)	**709.3** (Improvement **0.06**)	**6.38**	1st (by AIC)

The **Student t SWARCH-L(4, 2)** specification is the **only model investigated that has a better mean squared error** in forecasting u_t^2 than the constant variance specification.

B. Final Conclusion

The SWARCH specification offers a better statistical fit to the data and better forecasts. The results confirm:

1. **Persistence Source:** Most persistence is attributed to the long-lasting low-, moderate-, and high-volatility regimes.
2. **Leverage Effect:** Stock price decreases lead to a bigger increase in volatility than an equivalent increase.
3. **Distribution:** Fundamental innovations are better described by a **Student t distribution with low degrees of freedom** than by a Normal distribution.
4. **Serial Correlation:** Weekly stock returns are positively serially correlated.